

Modeling Profit Give-Back Risk in Algorithmic Trading via LSTM-Based Exit Decision Systems

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Abstract

This paper studies ... We examine ... Using ... data from ... to ..., we find that ...
The results suggest that ... This paper contributes to the literature by ...

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1 Introduction

Problem formulating: In quantitative trading, trend-following strategies especially moving average (MA) or weighted moving average (WMA) crossover strategies are widely used due to their structural simplicity and cross-market robustness. These strategies rely on MA crossover signals to initiate and reverse positions, effectively capturing medium- to long-term price trends. While entry rules have been extensively studied and well-established, exit timing optimization remains a critical bottleneck limiting strategy performance.

Traditional trend-following approaches typically exit positions only when an opposite crossover signal occurs, a mechanism that suffers from inherent lag. Although it captures full trend cycles, it often fails to lock in accumulated unrealized profits during late-stage trends or range-bound markets, leading to profit drawdowns. Existing solutions, such as fixed take-profit/stop-loss rules or trailing stops, rely on empirically calibrated parameters, struggle to adapt to dynamic market conditions, and lack data-driven frameworks for early exit decisions.

To systematically characterize this problem, we conduct an empirical analysis of the trading paths of traditional WMA strategies in the cryptocurrency market and identify a salient empirical fact that has not been sufficiently discussed: the vast majority of trades once achieved positive unrealized returns during the holding period, but only a small proportion eventually closed with positive returns. Specifically, in the dataset examined in this study, around 90% of trades were in profit at some point, whereas only about 30% ended as profitable trades (see in Table 1). Whereas the profit path (see in Figure 1) shows a highly consistent dynamic structure: in the early stage of the holding period, profits increase monotonically over time; in the middle-to-late stage, profits reach a peak corresponding to MFE; and in the later stage, a significant drawdown occurs. This finding suggests that the main bottleneck of trend-following strategies is not the absence of profit opportunities, but the lack of an effective mechanism for profit locking and risk control.

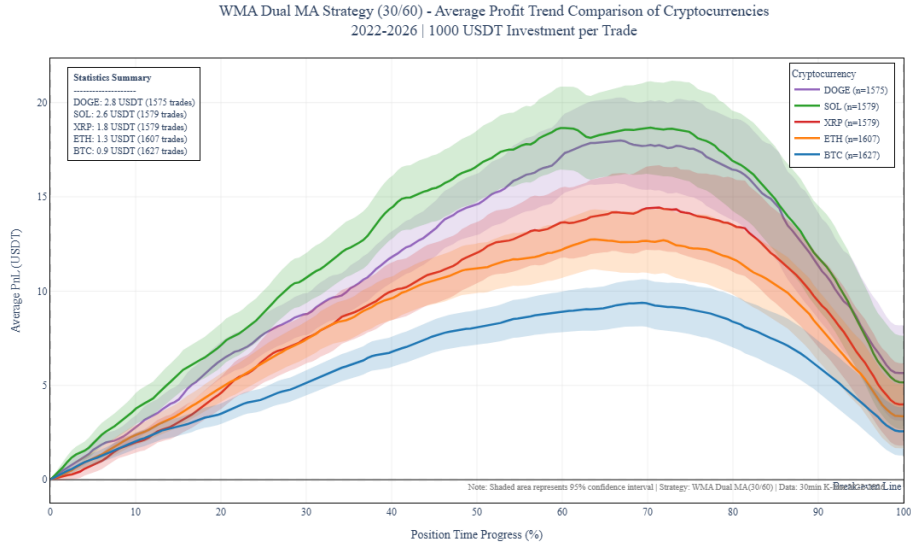
Based on the above results, this paper formalizes the phenomenon as unrealized profit drawdown risk and models it as a sequential decision-making problem: during the holding period, how can one determine, based on the current market state and position-related characteristics, whether continued holding will lead to a significant future drawdown in returns, and thereby decide whether the position should be exited early. Unlike traditional approaches that focus on price prediction or trend identification, this paper emphasizes the dynamic evolution of the profit path and its associated risk structure.

We propose an LSTM-based early-exit decision framework that reformulates the exit problem as a sequence classification task. The framework is designed to be embedded as a data-driven decision layer atop traditional trend-following strategies, learning the optimal exit boundary by balancing future returns and potential drawdowns. Backtesting

Table 1: Statistical Results of Trading Paths

Asset	Total Trades	MFE > 0 Trades	MFE > 0 Ratio	Final PnL > 0 Trades	Final PnL > 0 Ratio	Average MFE (USDT)	Average Final PnL (USDT)	Total PnL (USDT)
DOGE	1575	1371	87.05	566	35.94	36.75	2.79	4389.06
SOL	1579	1371	86.83	551	34.90	38.79	2.60	4103.44
XRP	1579	1359	86.07	552	34.96	30.58	1.77	2789.82
ETH	1607	1404	87.37	541	33.67	26.04	1.34	2152.53
BTC	1627	1399	85.99	536	32.94	19.02	0.91	1486.19

Note: MFE refers to Maximum Favorable Excursion, and PnL denotes Profit and Loss. All statistical results are quoted in USDT, and all ratio indicators are presented as percentages (%). The 30-minute frequency cryptocurrency trading data utilised in this paper is retrieved from the public API of Binance, spanning the sample period from January 1, 2022 to 2026. Within the adopted WMA strategy, the lengths of the long and short trading k-line windows are set to 60 and 30, respectively.



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Figure 1: Average Profit Paths Across Cryptocurrencies

Note: This figure illustrates the average trade profit corresponding to each cryptocurrency asset. The horizontal axis denotes the holding duration, while the vertical axis represents profit measured in USDT. The shaded area corresponds to the 95% confidence interval.

on cryptocurrency market data demonstrates that the proposed method significantly improves key performance metrics while overall returns remain stable.

The remainder is organized as follows. Section 2 reviews the related literature in this field.

2 Related Work

Quantitative trading research has long focused on identifying price regularities and transforming them into implementable strategies. The literature has evolved from traditional technical analysis to machine learningbased prediction and, more recently, to reinforcement learningbased sequential decision making. While extensive studies have explored entry signals and price forecasting, optimal exit timing especially for mitigating unrealized profit drawdown remains understudied.

2.1 Trend-following Strategies

Trend-following represents one of the most classical paradigms in quantitative trading, which exploits price inertia through technical indicators to identify and trade in line with prevailing trends. Moving-average-based methods are the most representative implementations, including the simple moving average (SMA), weighted moving average (WMA), exponential moving average (EMA), and short-long moving-average crossover rules. [Hunter \(1986\)](#) provided an early systematic discussion of the exponentially weighted moving average, emphasizing the importance of recent information in time-series analysis. [Hansun \(2013\)](#) further discussed the evolution and improvement of traditional moving-average tools.

As research developed, scholars began to examine the effectiveness of moving-average rules from theoretical and empirical perspectives rather than treating them as purely empirical tools. [Chiarella et al. \(2006\)](#) analyzed the mechanism of moving-average rules from a dynamic market perspective, noting that window length and trading feedback significantly affect price dynamics. [Ellis and Parbery \(2005\)](#) compared adaptive and simple moving-average strategies, showing that dynamic parameter adjustment improves adaptability. [Marshall et al. \(2017\)](#) compared time-series momentum and moving-average rules, highlighting differences in signal construction and return sources.

Overall, trend-following strategies have evolved from fixed-parameter rules toward adaptive and diversified structures. Traditional technical analysis has not become obsolete with the rise of machine learning; instead, it provides important features for intelligent trading models. For instance, [Ayala et al. \(2021\)](#) showed that technical strategies are increasingly optimized and integrated via machine learning, enabling a transition from rule-driven to data-driven frameworks.

2.2 Machine Learning in Trading

The application of machine learning has significantly transformed the paradigm of quantitative trading. Unlike traditional methods that rely on predefined rules, machine learning automatically captures nonlinear mappings between market data and trading outcomes. Early studies such as [Trippi and DeSieno \(1992\)](#) applied neural networks to index futures trading, while [Dempster and Jones \(2001\)](#) and [Dempster and Leemans \(2006\)](#) introduced genetic programming and reinforcement learning, reflecting a shift from manual rule design to automated decision making.

Supervised learning in trading is primarily divided into classification and regression. Classification methods predict price directions or trading signals [Shen et al. \(2012\)](#); [Gerlein et al. \(2016\)](#); [Zhong and Enke \(2019\)](#), whereas regression methods focus on forecasting price levels and returns [Adegboye and Kampouridis \(2021\)](#).

LSTM has become a milestone in financial time-series modeling due to its ability to capture long-term dependencies. [Nelson et al. \(2017\)](#) and [Fischer and Krauss \(2018\)](#) demonstrated its strong performance in direction prediction. [Borovkova and Tsiamas \(2019\)](#) extended LSTM to high-frequency data, and [Mehtab et al. \(2020\)](#); [Ghosh et al. \(2022\)](#) further developed hybrid structures combining LSTM with other models.

Recently, machine learning research has shifted from pure prediction accuracy to strategy-oriented performance. [Sebastião and Godinho \(2021\)](#) highlighted that in-sample precision does not guarantee trading profits. [Han et al. \(2023\)](#) linked label design more closely with trading system construction, indicating a trend toward profit-driven and strategy-aware modeling.

2.3 Exit Strategy Optimization

Compared with entry signal research, exit strategies directly determine profit realization and risk control but have received far less attention. Exit decisions affect not only trade-level PnL but also drawdown management and risk exposure.

Early exit research focused on fixed take-profit and stop-loss rules. [Leung and Li \(2015\)](#) incorporated stop-loss constraints into mean-reversion strategies, emphasizing the impact of exit boundaries on optimality. [Lo and Remorov \(2017\)](#) showed that stop-loss performance depends heavily on market states. [Vezeris et al. \(2018\)](#) found that dynamic volatility-based thresholds outperform fixed ones.

Trailing stop-loss rules further enable profit locking by adjusting exit levels with favorable price movements. [Glynn and Iglehart \(1995\)](#) provided early theoretical analysis, while [Dai et al. \(2021\)](#) and [Leung and Zhang \(2021\)](#) demonstrated their ability to reduce downside risk and formalized trailing stops as optimal stopping problems.

Reinforcement learning has recently emerged as a tool for sequential exit decisions. [Kim and Kim \(2019\)](#), [Tsantekidis et al. \(2021\)](#), and [Wang et al. \(2021\)](#) applied deep

reinforcement learning to optimize trading boundaries and exit quality. However, such methods remain more focused on overall strategy optimization rather than fine-grained exit timing for individual trades.

Despite progress, research on exit optimization remains limited. [Hwang et al. \(2023\)](#) noted that exit rules strongly influence strategy efficiency, while [Sivri et al. \(2023\)](#) and [Shawky \(2023\)](#) represent early attempts to use machine learning for stop-loss and take-profit calibration. The field still lacks systematic, data-driven frameworks for dynamic exit and drawdown control.

2.4 Research Gaps

Existing literature has achieved substantial progress in trend identification, price prediction, and automated trading. However, critical limitations remain in exit optimization, particularly for controlling unrealized profit drawdowns.

First, traditional trend-following strategies rely on fixed parameters and lack robustness to volatile or structurally changing markets.

Second, exit research remains underdeveloped relative to entry signal modeling. Studies on trailing exits, dynamic take-profit mechanisms, and integrated exit optimization are especially limited.

Third, most studies treat entry, holding, and exit as separate components rather than building a unified decision framework. Poor exit design can erode accurate predictions through profit drawdowns, unnecessary stop-losses, and excessive transaction costs.

In summary, while existing research provides a solid foundation for trend following and machine learning-based prediction, data-driven dynamic exit optimization especially for profit protection and drawdown control remains an underexplored area with clear theoretical and practical value.

3 Methodology

3.1 Framework Overview

This paper proposes a hybrid decision-making framework that combines the signal identification capability of traditional trend-following strategies with the nonlinear risk-control capability of deep learning models. The framework consists of three key components: baseline strategy layer, LSTM decision layer, and exit execution mechanism. The baseline strategy layer uses weighted moving average (WMA) crossover signals as the logical starting point, and is responsible for capturing market trends and generating initial trading signals. The LSTM decision layer serves as an intelligent risk-management module embedded on top of the baseline strategy. By monitoring real-time market-state

features and position-context information, it dynamically evaluates the risk of unrealized profit drawdown.

The core operating logic of the system is a two-layer collaborative mechanism. When the baseline strategy layer triggers an entry signal according to the moving-average crossover rule, the system enters the holding state. At this point, the LSTM decision layer is activated simultaneously. Through nonlinear mapping of the input feature sequence, it outputs in real time the conditional probability of whether an early closing operation should be executed at the current time. If the predicted probability exceeds a preset decision threshold, the system will execute an exit operation before the traditional trend-reversal signal appears, so as to lock in profits and avoid or mitigate potential large drawdown. From a system-architecture perspective, this model constitutes a data-driven decision layer superimposed on the traditional rule-based logic, thereby significantly improving the refined management capability of the strategy during the holding period.

3.2 Baseline Strategy

To ensure the sensitivity and representativeness of the empirical analysis, this paper selects a weighted moving average (WMA) dual moving-average system as the baseline strategy. Compared with a simple moving average (SMA), WMA assigns greater weights to more recent price observations, enabling it to respond more quickly to price changes and to exhibit greater flexibility in trend identification. The strategy logic adopted in this study follows a standard moving-average crossover rule. In terms of entry rules, the system uses the crossover logic of a 30-period fast moving average and a 60-period slow moving average. When the fast moving average crosses above the slow moving average from below, that is, when a golden cross is formed, a long-entry signal is executed. Conversely, when the fast moving average crosses below the slow moving average from above, that is, when a death cross is formed, a short-entry signal is executed. In terms of traditional exit rules, the baseline mechanism used for comparison relies only on reverse trading signals; that is, a position is closed only when an opposite signal to the current position direction appears. Although this mechanism can help capture relatively complete trend intervals, it also faces significant unrealized profit drawdown risk in the later stage of a trend, which constitutes the core motivation for introducing the LSTM decision layer in this paper.

3.3 Problem Formulation

In the hybrid decision-making framework proposed in this paper, the LSTM decision layer does not directly undertake the task of predicting trend direction. Instead, it is defined as a dynamic exit decision model for the holding period. In other words, the baseline strategy layer has already determined the trading direction according to the

WMA crossover rule, while the core objective of the LSTM model is to judge whether the current unrealized profit faces a relatively high future drawdown risk during the holding process.

Let the market feature vector at time t be denoted by $\mathbf{x}_t$. For any trading sample that is in the holding state, the model takes the feature sequence over the past k time steps as input:

$$\mathbf{X}_{t-k:t} = \{\mathbf{x}_{t-k}, \mathbf{x}_{t-k+1}, \dots, \mathbf{x}_t\} \quad (1)$$

The output of the LSTM decision layer is defined as the conditional probability of executing an early closing operation at the current time:

$$p_t = P(exit \mid \mathbf{X}_{t-k:t}) \quad (2)$$

This probability can be regarded as a probabilistic characterization of “unrealized profit drawdown risk.” When p_t is high, it indicates that the model believes that, if the position continues to be held, the current unrealized profit is more likely to experience a future drawdown. Conversely, when p_t is low, it suggests that continuing to hold the position may still have a higher expected value than exiting immediately.

Therefore, the problem in this paper can be viewed as a sequential binary classification problem: given the current and historical market-state sequence $\mathbf{x}_{t-k:t}$, the model learns to determine whether the position should be actively exited before the reverse signal of the baseline strategy appears. In this way, the LSTM layer is transformed from a traditional “direction prediction model” into a “profit protection and risk control model.”

3.4 Feature Engineering

To enable the LSTM decision layer to perceive both the market environment and the current trading state, this paper constructs an input feature system jointly composed of market features and position-context features, providing the informational basis required for the model to identify future unrealized profit drawdown risk.

3.4.1 Market Features

When selecting market features, this paper follows the approaches of [Fischer and Krauss \(2018\)](#), [Huang et al. \(2024\)](#), and [AlKhatib et al. \(2022\)](#), and jointly considers price trends, return-generating ability, candlestick patterns, volatility conditions, and trading activity. The detailed definitions of the variables are shown in Table 2.

Table 2: Market Features

Category	Feature	Description
Trend	wma30_rel, wma60_rel	Deviation of the closing price from the moving averages
Trend	wma_spread	Deviation of the 30-day moving average from the 60-day moving average
Trend	wma30_slope, wma60_slope	First-order differences of the moving averages
Return	log_ret, oc_ret	Log return and open-close ratio
Pattern	hl_range, shadow_imbalance	Price range and upper-lower shadow imbalance
Volatility	atr14_ret	ATR(14) divided by the closing price
Trading Activity	vol_z20, taker_buy_ratio	Trading volume Z-score and active buy ratio

3.4.2 Position Context Features

In addition to market features, the key innovation of this paper lies in the introduction of position-context features. Traditional technical indicators usually describe the market itself, but cannot represent the current trading state. However, for the early take-profit problem, the decision implications of the same market trend may differ across different stages of the holding period. For example, the drawdown immediately after opening a position and the drawdown after a large unrealized profit are not equivalent in terms of risk implications. Therefore, this paper represents the position state as model input. The detailed definitions of the variables are shown in Table 3.

Table 3: Position Context Features

Feature	Description
age	Number of bars for which the current position has been held
pnl_dir	Unrealized profit or loss in the direction of the position
mfe	Maximum favorable excursion
retrace	Drawdown from the maximum favorable excursion
dir	Position direction, where “+1” denotes a long position and “-1” denotes a short position

By introducing the above position-context information, the LSTM model no longer makes decisions solely based on market states, but is able to make decisions according to the dynamic profit and loss condition of the current trade, the holding duration, and other information. This allows the model to more closely approximate the risk-management logic of real trading.

3.5 Label Construction

To train the LSTM decision layer, this paper needs to construct a supervisory label indicating whether an early exit should be executed for each time point in the holding state. This paper adopts a value-based labeling method based on the trade-off between future returns and future drawdowns, so that the labels can more closely reflect unrealized profit drawdown risk.

For a trade generated by the baseline strategy, let its entry time be t_0 and the original exit time of the baseline strategy be T . At any holding time t , if the position is held until the exit point given by the baseline strategy, the future incremental return that may still be obtained relative to the current time is defined as:

$$future_gain_t = pnl_T - pnl_t \quad (3)$$

where pnl_t denotes the unrealized profit adjusted by the current position direction, and pnl_T denotes the profit when the baseline strategy eventually exits.

At the same time, if the position continues to be held from the current time, the price may experience adverse fluctuations along the future path. This paper defines the future adverse drawdown as $future_drawdown_t$, which is used to measure the maximum adverse price movement that the trade may encounter between the current time and the baseline exit time. Based on this, this paper constructs the value evaluation function:

$$value_score_t = future_gain_t - \lambda \cdot future_drawdown_t \quad (4)$$

where λ is the risk-penalty coefficient, used to reflect investors' aversion to risk. When λ is larger, the label construction process places greater emphasis on drawdown risk; conversely, it places more emphasis on the future returns that may be obtained by continuing to hold the position.

This paper uses the value score as a proxy indicator for unrealized profit drawdown risk. If $value_score_t < 0$, it indicates that the potential return from continuing to hold the position from the current time is insufficient to compensate for the adverse drawdown that may occur in the future. In this case, the sample should be labeled as an early exit:

$$y_t = \begin{cases} 1, & \text{if } value_score_t < 0 \\ 0, & \text{if } value_score_t \geq 0 \end{cases} \quad (5)$$

where $y_t = 1$ indicates that an early closing operation should be executed at the current time, while $y_t = 0$ indicates that the position should continue to be held. In this way, the model's learning objective is no longer to predict the price direction, but to identify whether the current position is exposed to relatively high future drawdown risk.

To prevent the model from being disturbed by noise in the early stage after opening

a position, this paper sets a minimum holding-period constraint. A sample is included in the label construction process only when the holding time of the trade exceeds 90 minutes and the current unrealized profit exceeds 0.5%. This treatment ensures that the LSTM decision layer mainly serves the exit management after an unrealized profit has already emerged, rather than interfering with the initial entry logic of the baseline strategy.

3.6 LSTM Model

This paper adopts a Long Short-Term Memory (LSTM) network as the early-exit decision model. LSTM is suitable for capturing dynamic dependence in financial time series and can, to a certain extent, capture the nonlinear relationship between trend evolution and position-state changes.

The model input is a fixed-length historical feature matrix:

$$\mathbf{X}_{t-63:t} \in \mathbb{R}^{64 \times 17} \quad (6)$$

where 64 denotes the length of the time window, and 17 denotes the feature dimension contained in each time step.

The LSTM model recursively encodes the input sequence and uses the hidden state of the last time step as a comprehensive representation of the current holding state. The hidden state is then mapped into a one-dimensional output through a fully connected layer and converted into a probability form through the Sigmoid function:

$$p_t = \sigma(\mathbf{w}\mathbf{h}_t + b) \quad (7)$$

where $\mathbf{h}_t$ denotes the hidden state of the LSTM at the current time, $\mathbf{w}$ and b denote the weight and bias of the fully connected layer, respectively, and $\sigma(\cdot)$ denotes the Sigmoid activation function. The final output is interpreted as the probability of early exit at the current time, namely:

$$p_t = P(y_t = 1 \mid \mathbf{X}_{t-63:t}) \quad (8)$$

When p_t is close to 1, it indicates that the model believes the future unrealized profit drawdown risk is relatively high and that the current position should be closed early. When p_t is close to 0, it indicates that continuing to hold the position is more reasonable.

3.7 Decision Rule

In the backtesting stage, the LSTM decision layer operates as an additional exit module on top of the baseline strategy. The baseline strategy still generates trading directions and initial entry signals, while the LSTM model participates in decision-making

only when an existing position is being held.

Specifically, when the system executes a WMA crossover rule based on the baseline strategy and establishes a position, for each new candlestick, the market features and position-context features are updated, and the feature sequence of the most recent 64 time steps is input into the LSTM model to obtain the early-exit probability p_t .

The exit rule in this paper can be expressed as:

$$Exit_t = \begin{cases} 1, & p_t > \theta \\ -1, & \text{opposite WMA crossover occurs} \\ 0, & \text{else} \end{cases} \quad (9)$$

where θ is the decision threshold for early exit, determined through grid search, and the results are shown in Table 4. When $p_t > \theta$, it indicates that the model judges the current position to face a relatively high future unrealized profit drawdown risk, and therefore the system will actively close the position before the reverse crossover signal of the baseline strategy appears. If the LSTM does not trigger an early exit, the trade continues to be managed according to the original moving-average reverse crossover rule.

Table 4: Backtesting Results under Different LSTM Decision Thresholds

Threshold θ	Number of LSTM Exits	LSTM Exit Ratio (%)	Cumulative Return (%)	Win Rate (%)	Sharpe Ratio	Average Holding Bars	Maximum Drawdown
Baseline	0	0.0	+1.2807	36.7	2.252	48.9	0.2782
0.30	142	68.6	+0.4161	68.6	3.723	11.2	0.1045
0.40	141	68.1	+0.4511	68.1	3.842	11.6	0.1390
0.50	121	58.5	+0.8222	64.3	3.960	18.6	0.1336
0.55	58	28.0	+0.9526	51.2	2.602	34.2	0.1351
≥ 0.60	0	0.0	+1.2807	36.7	2.252	48.9	0.2782

Through the above decision rule, this paper forms a two-layer exit mechanism: the baseline strategy provides trend-reversal exits, while the LSTM layer provides risk-warning-based early exits. The former ensures that the strategy still follows trend-following trading logic, while the latter improves the responsiveness to unrealized profit drawdown risk during the holding period. This mechanism enables the strategy to no longer passively wait for the moving-average reverse crossover, but instead to lock in profits in advance when drawdown risk rises, thereby improving strategy performance.

4 Experimental Design

4.1 Dataset

To objectively evaluate the generalization ability of the LSTM-based early-closing decision framework in dynamic market environments, this study selects the cryptocurrency market as the empirical setting. The cryptocurrency market features 24/7 continuous trading, high volatility, and pronounced trend persistence, making it an ideal setting for testing whether the enhanced strategy can effectively improve the traditional trend-following model.

This study selects five mainstream cryptocurrency trading pairs as the experimental assets: BTC/USDT, ETH/USDT, SOL/USDT, XRP/USDT, and DOGE/USDT. These assets cover different market capitalizations, liquidity characteristics, and price fluctuation patterns, thereby allowing the robustness of the model across assets to be evaluated.

The raw market data used in the experiment range from January 1, 2022 to March 12, 2026. This interval fully covers multiple bull-bear cycles in the market, enabling the model to be tested across different market regimes, including trend continuation, range-bound fluctuation, and sharp drawdown. In this paper, 30-minute candlestick data are used for modeling. This time granularity ensures the reliability of trend signals while providing sufficient sequence depth for the LSTM model, so as to capture the dynamic evolution of micro risk structures during the holding period.

This study strictly follows the chronological order principle for predictive samples, and divides the dataset into the training set, validation set, and testing set, with proportions of approximately 70%, 15%, and 15%, respectively. The specific sample sizes and corresponding time intervals are shown in Table 5.

Table 5: Dataset Split

Stage	Proportion	Sample Size	Time Interval	Research Function
Training Set	70%	29,250	2022/01/02–2025/01/06	Model parameter learning and weight optimization
Validation Set	15%	6,268	2025/01/06–2025/08/05	Hyperparameter tuning and overfitting prevention
Testing Set	15%	6,269	2025/08/05–2026/03/11	Out-of-sample evaluation and robustness test

4.2 Data Cleaning

This paper first conducts basic cleaning on the raw candlestick data, including uniformly parsing the time field into UTC time format, converting fields such as prices and

trading volume into numerical types, and sorting the data in chronological order. Subsequently, anomaly auditing is performed on the raw sequence to identify missing candlesticks, zero-volume candlesticks, and extreme fluctuation samples, with corresponding flags constructed for each type of anomaly. Since the LSTM model adopts a fixed-length historical window for modeling, this paper further constructs a context-contamination flag using a rolling-window method, and removes samples whose historical windows contain bad data from the training dataset.

4.3 Comparison Methods

To examine whether the proposed Long Short-Term Memory (LSTM)-based early-closing decision framework can further improve performance on top of traditional trading strategies, this paper sets up several representative benchmark methods for comparative experiments. Specifically, the proposed LSTM early-closing strategy is compared with the WMA baseline strategy, the fixed take-profit strategy, and the trailing take-profit strategy, in order to evaluate its effectiveness in enhancing returns, controlling drawdowns, and improving risk-adjusted returns.

First, the WMA baseline strategy serves as the fundamental trend-following strategy in this paper, and is used to characterize the original strategy performance without introducing any additional early-closing mechanism. This method mainly relies on 30-day and 60-day weighted moving average signals for opening and closing positions, thereby providing a basic reference for evaluating whether subsequent exit-optimization methods are effective.

Second, the fixed take-profit strategy is a simple and commonly used rule-based exit method. It triggers position closing once the holding return reaches a fixed threshold, which is set to 10% in this paper. The advantage of this method lies in its clear rules and stable execution. However, its limitation is that it cannot adjust dynamically according to market conditions, which may lead to premature profit taking or failure to avoid profit drawdowns in a timely manner.

Third, the trailing take-profit strategy is used to represent a more dynamic traditional risk-control method. When the holding return reaches a preset threshold, which is set to 3% in this paper, the strategy enters a take-profit monitoring state. Thereafter, if the current holding return falls by more than a preset threshold from the highest unrealized profit level reached during the holding period, which is set to 10% in this paper, the position is closed. Compared with the fixed take-profit strategy, the trailing take-profit strategy can preserve the potential for further gains during trending markets to some extent while controlling the drawdown of unrealized profits. Therefore, it serves as an important benchmark for testing whether the proposed method has stronger dynamic risk-identification capability.

In terms of experimental design, this paper applies the above strategies uniformly to multiple mainstream cryptocurrency trading pairs, including BTC/USDT, ETH/USDT, SOL/USDT, XRP/USDT, and DOGE/USDT, and compares their performance across different assets. Through cross-asset experiments, the robustness of the proposed method can be further examined.

4.4 Evaluation Metrics

To comprehensively evaluate the performance of different strategies and models in the early-closing task, this paper constructs an evaluation metric system from five dimensions: profitability, risk-adjusted return, drawdown control, trading stability, and holding efficiency. Specifically, the metrics include cumulative return, Sharpe ratio, maximum drawdown, win rate, and average holding time.

First, cumulative return is used to measure the overall profitability of the strategy during the testing period. This metric reflects the growth of the initial capital after the complete backtesting period and is the most direct indicator for evaluating strategy return performance. It is calculated as follows:

$$Ret = \frac{V_T - V_0}{V_0} \quad (10)$$

where V_0 denotes the initial capital size, and V_T denotes the capital size at the end of the backtesting period. A higher cumulative return indicates stronger overall profitability during the out-of-sample testing period.

Second, the Sharpe ratio is used to measure the excess return obtained by the strategy per unit of risk taken. Compared with a simple return metric, the Sharpe ratio further considers return volatility, and therefore can better reflect the risk-adjusted return performance of the strategy. It is calculated as follows:

$$SharpeR = \frac{R_p - R_f}{\sigma_p} \quad (11)$$

where R_p denotes the strategy return, R_f denotes the risk-free rate, and σ_p denotes the standard deviation of strategy returns. In high-frequency or short-period cryptocurrency backtesting, since each holding period is relatively short, the risk-free rate can be approximately set to zero in this paper. A higher Sharpe ratio indicates that the strategy can obtain higher returns under the same level of risk.

Third, maximum drawdown is used to measure the maximum capital loss that the strategy may suffer during the backtesting period, and is an important indicator for evaluating the risk-control capability of the strategy. It is calculated as follows:

$$MDD = \max_{t \in [0, T]} \frac{V_t^{peak} - V_t}{V_t^{peak}} \quad (12)$$

where V_t denotes the net asset value at time t , and V_t^{peak} denotes the historical highest net asset value up to time t . A smaller maximum drawdown indicates stronger capital-protection capability under adverse market fluctuations. For an early-closing strategy, if the model can identify profit drawdown risk in a timely manner, it should reduce the maximum drawdown to some extent.

Fourth, the win rate is used to measure the stability of trading outcomes, and represents the proportion of profitable trades among all trades. It is calculated as follows:

$$WinRate = \frac{N_{win}}{N_{total}} \quad (13)$$

where N_{win} denotes the number of profitable trades, and N_{total} denotes the total number of trades. A higher win rate indicates that the strategy generates profitable trades more frequently. However, the win rate alone cannot represent the overall quality of a strategy, because a high-win-rate strategy may still produce poor overall returns due to a small number of large losses. Therefore, this paper analyzes it jointly with cumulative return, Sharpe ratio, and maximum drawdown.

Finally, average holding time is used to measure the average duration from opening to closing a position, reflecting the impact of different closing mechanisms on the holding period. It is calculated as follows:

$$AvgHT = \frac{1}{N} \sum_{i=1}^N (t_i^{exit} - t_i^{entry}) \quad (14)$$

where t_i^{entry} and t_i^{exit} denote the opening time and closing time of the i -th trade, respectively, and N denotes the total number of trades. Average holding time can reflect whether a strategy tends to exit too early or hold positions for a long time. For the LSTM-based early-closing framework proposed in this paper, this metric helps determine whether the model can avoid subsequent drawdown risk in a timely manner while preserving the profit potential of trend-following trades.

- 5 Results
- 6 Discussion
- 7 Robustness Checks
- 8 Conclusion
- 9 Future Work

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A Additional Tables

Table 6: Variable Definitions

Variable	Definition
$Flow_t^{5d}$	Five-day change in stablecoin market capitalization
$\Delta \ln Flat_t$	Growth rate of fiat-reserve-backed stablecoin market capitalization
$\Delta \ln Coll_t$	Growth rate of collateralized stablecoin market capitalization
Δy_t	Change in Treasury yield
X_t	Vector of control variables

B Additional Figures

Additional figures can be placed here.